CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The rapid growth of digital technology has transformed the way people plan their travel and accommodation. Online hotel booking systems have become highly popular due to their convenience and accessibility. Users can explore different hotels, compare prices, and make reservations easily. However, many existing platforms lack proper guidance and interaction. This often leads to confusion while selecting the best option. Therefore, there is a need for a smarter and more interactive booking system.

Traditional hotel booking systems mainly focus on displaying information without providing intelligent assistance. Users are required to manually search, filter, and decide based on limited details. This process can be time-consuming and sometimes overwhelming. Especially for first-time users, the absence of guidance becomes a major issue. As a result, users may not always make the best choice. This highlights the importance of integrating advanced technologies into booking platforms.

Artificial Intelligence (AI) has emerged as a powerful solution to improve user experience in various domains. In the hospitality industry, AI can be used to provide personalized recommendations and instant support. Technologies like Machine Learning and Natural Language Processing enable systems to understand user behavior. These technologies help in analyzing user preferences and providing accurate suggestions. AI-based systems can also learn and improve over time. This makes them more efficient and user-friendly.

The concept of an AI avatar takes user interaction to the next level by simulating human-like communication. The avatar can interact with users through voice or text and guide them step by step. It can answer queries, suggest hotels, and assist in completing bookings. This creates a more engaging and comfortable experience for users. The avatar acts as a virtual assistant available at all times. This reduces dependency on human customer support services.

In addition to interaction, personalization is a key feature of the proposed system.The AI avatar analyzes user inputs such as budget, location, ratings, and preferences. Based on this data, it recommends the most suitable hotel options. This saves time and helps users make better decisions. The system can also provide real-time updates on room availability and pricing. Such features enhance the overall efficiency of the booking process.

This project focuses on developing an AI Avatar Assistance system for online hotel room booking. The system aims to provide a smart, interactive, and user-friendly platform. It integrates AI technologies to simplify the booking process and improve user satisfaction. The proposed solution reduces manual effort and enhances decision-making. It also ensures secure and reliable booking with multiple payment options. Overall, this system represents a modern approach to hotel booking services

1.2 OVERVIEW

The AI Avatar Assistance for Online Hotel Room Booking system is an advanced solution designed to improve the overall hotel reservation experience. It combines Artificial Intelligence with an interactive avatar to guide users throughout the booking process. The system provides a user-friendly platform where customers can search for hotels, compare options, and complete bookings .

By integrating intelligent features, the system reduces complexity and enhances convenience. It aims to make hotel booking faster, smarter, and more engaging.

The core component of this system is the AI-powered avatar, which acts as a virtual assistant. It interacts with users using text or voice-based communication, making the system more intuitive. The avatar understands user queries and provides relevant responses in real time. It can suggest hotels based on user preferences such as location, budget, and amenities. This interactive approach helps users feel more comfortable and confident while making decisions.

Another important aspect of the system is its recommendation capability. Using AI techniques like Machine Learning, the system analyzes user behavior and preferences. Based on this analysis, it provides personalized hotel suggestions that match the user's needs. This reduces the time spent searching and filtering options manually. It also improves user satisfaction by offering accurate and relevant results.

The system also includes a seamless booking and payment process. Users can easily select rooms, confirm their bookings, and make payments through secure payment gateways. Features such as real-time availability, booking confirmation, and notification updates are included. These functionalities ensure that users have a smooth and reliable booking experience without errors or delays.

Overall, the AI Avatar Assistance system provides a modern and innovative approach to hotel booking. It enhances user interaction, improves decision-making, and automates the booking process. The system is designed to meet the needs of today's digital users by offering speed, accuracy, and convenience. This project demonstrates how AI can transform traditional booking systems into intelligent and interactive platforms.

1.3 OBJECTIVE

- The main objective of this project is to design and develop an AI-based avatar that can assist users in booking hotel rooms. The avatar should interact with users in a natural and human-like way. It must support both text and voice communication to improve usability. The system should understand user queries and provide accurate responses. This makes the booking process more interactive and efficient. Overall, it enhances user experience through intelligent assistance.
- Another objective is to provide personalized hotel recommendations based on user preferences. The system should analyze factors such as budget, location, ratings, and amenities. Using AI techniques, it must suggest the most suitable hotel options. This reduces the time spent on searching and filtering manually. It also helps users make better decisions with ease. Personalization improves satisfaction and effectiveness of the system.
- The project also aims to simplify and automate the hotel booking process. Users should be able to search, select, and book rooms without any difficulty. The AI avatar will guide users step by step throughout the process. Automation reduces manual effort and minimizes human errors. It also speeds up the booking process and ensures accuracy. This provides a smooth and hassle-free user experience.
- Finally, the system focuses on ensuring secure and reliable transactions. It should support multiple payment methods such as UPI, debit/credit cards, and digital wallets. The system must protect user data and transaction details effectively. Features like booking confirmation and notifications should be included. This builds trust and reliability among users. Security plays a vital role in the success of the platform

CHAPTER 2

LITERATURE SURVEY

2.1 Jalel Ktari, Tarek Frikha, Majdi Rekik, Oussema Gassara, Marwen Boujelben and Nader Ben Amor "Decentralized Hotel Reservations: Blockchain Integration for Incentivized Data Sharing and Smart Contracts" 2024 IEEE 4th International Maghreb Meeting of the Conference on Sciences and Techniques of Automatic Control and Computer Engineering (MI-STA),2024

The paper describes a decentralized infrastructure that uses blockchain technology to securely manage hotel reservations and user profiles. The network leverages smart contracts to foster data exchange among vendors of travel services, and participants are rewarded with digital tokens. Financial transactions are facilitated via a unique wallet identification connected to a cryptocurrency server. With tools like as Solidity, Ethereum, Node.js, React, and MongoDB, the system provides safe and efficient hotel reservation management, as seen by quick response times and low transaction fees

2.2 Shukla R Vernekar, Lava Kumar and Gauri Kalnoor "Hotel Reservation App Integrated with Docker and Jenkins" 2024 International Conference on Emerging Technologies in Computer Science for Interdisciplinary Applications (ICETCS),2024

This article discusses the seamless integration of Docker with Jenkins in order to boost the productivity and scalability of a hotel booking application. The integration simplifies the Continuous Integration/Continuous Deployment (CI/CD) process by using Docker containerization and Jenkins automation to produce coherent environments.

Key features include OTP login, JSON Web Token (JWT) for secure sessions, and complete room details, ensuring a dependable and secure booking experience.

2.3 Ioana Udrea, Cristian Gabriel Alionte, Georgeta Ionaşcu, Tudor Cătălin Apostolescu "New research on People Counting and Human Detection" 2021 13th International Conference on Electronics, Computers and Artificial Intelligence (ECAI),2021.

This work investigates methods for counting individuals and recognizing persons in a room, with a specific emphasis on analyzing infrared pictures using computer vision in artificial intelligence (AI). The study contrasts existing systems that count people using PIR and ultrasonic sensors with a person detection system based on the Grid-EYE camera, which has an 8x8 pixel resolution. Because of the Grid-EYE camera's limitations, the authors recommend using a higher resolution Lepton infrared camera (80x60 pixels). Both cameras' results are compared, and conclusions are made on the usefulness of various people-counting techniques and possible uses for room occupancy data.This study investigates approaches for counting individuals and recognizing persons in a room, with an emphasis on computer vision in Artificial Infrastructure Yet, further study is needed to determine the influence of variables like seasonality and location on price

2.4 N. S. Lestari, D. Rosman, A. Faridi, B. E. Sukma, S. Rokhmah and A. Gunawan "The Effect of Technology Readiness and Customers" Acceptance on Online Hotel Booking Intention"2023 8th International Conference on Business and Industrial Research (ICBIR),2023.

This study looks at how customers' technological resilience impacts their acceptance of online hotel reservation systems via hotel websites, and how it affects their booking intentions.

Data was collected via snowball sampling in an online survey with 190 respondents in Jakarta. According to Partial Least Squares - Structural Equation Modeling (PLS-SEM), technological readiness has a favorable influence on perceived utility, convenience of use, and joy of hotel booking websites. This indicates that easier navigation for people with no prior knowledge is impotant to attract consumers

2.5 J. Deepthi and T. V. Gopal "Pre Encryption Data hiding techniques using Reserving Room Approach" 2022 IEEE International Conference for Women in Innovation, Technology & Entrepreneurship (ICWITE),2022.

The study puts forward a reversible data concealing approach that embeds data covertly into media (text, photos, and videos) while allowing complete retrieval of both the hidden data and the cover media. Pixel prediction serves to retrieve hidden data from encrypted photos and visible watermarks without loss of quality. Irrespective of the potential faults caused by altering encrypted data, error correcting codes verify that data extraction and image restoration are accurate. The recommended method beats earlier methods by having a greater embedding rate, resulting in better outcomes

CHAPTER 3

SYSTEM STUDY

3.1 EXISTING SYSTEM

- Users must manually search for hotels without intelligent assistance, making the process time-consuming and less efficient. There is also limited interaction due to the absence of AI or virtual assistants, so users do not receive proper guidance during booking. Overall, this reduces ease of use and convenience.
- The system lacks personalized recommendations based on user preferences, forcing users to select hotels without smart suggestions. Filtering and comparing hotel options also takes more time and can confuse users due to too many choices. This negatively impacts decision-making efficiency.
- The system does not provide real-time support, so users may face issues without immediate help. The booking process involves multiple manual steps, which increases complexity for users. As a result, the overall experience becomes slower and less smooth.
- There are higher chances of human errors during booking, which can lead to booking failures or wrong selections. Some systems also have weak security measures, creating risks during online payments. These issues reduce user confidence and trust.
- Users may not receive instant confirmation or timely updates after booking. This lack of communication reduces reliability and creates uncertainty. It further affects the overall satisfaction with the system.
- Overall, the system lacks automation, intelligence, and user-friendly features, making it less efficient. The absence of smart support and strong security reduces its effectiveness. These limitations clearly show the need for improvement in usability and trust.

3.1.1 DISADVANTAGES

- The system does not provide AI-based assistance to users, making the booking process less interactive and more difficult. There is also no personalized recommendation feature, so users may not easily find hotels that match their preferences. Overall, this reduces convenience and makes the experience less user-friendly.
- The manual search process takes more time as users need to apply filters and compare options on their own. The system also offers limited user interaction, which reduces engagement and overall satisfaction. As a result, users may find the process slow and less efficient.
- There is no real-time customer support available, so users may face issues without immediate assistance. The booking process is also complex and involves multiple steps, which can confuse users. This increases effort and reduces ease of use.
- Higher chances of human errors during booking can lead to incorrect reservations or booking failures. In addition, security issues in payment systems may create risks for user data and transactions. These problems reduce trust in the platform.
- Booking confirmations may be delayed or unclear, which further reduces reliability. Users may not receive proper updates after completing transactions. This creates uncertainty and affects overall satisfaction.
- Overall, the system lacks automation, intelligence, and strong support features. The absence of smart recommendations and real-time assistance makes it less efficient. These limitations highlight the need for significant improvements in usability and security.

3.2 PROPOSED SYSTEM

- The system uses an AI avatar to assist users in hotel booking by providing human-like interaction through text or voice. It offers personalized hotel recommendations based on user preferences. The platform has a simple and user-friendly interface for easy navigation
- Additionally, the system includes smart search and filtering options based on price, location, and amenities. The AI avatar provides real-time assistance and guides users throughout the booking process. This ensures a smooth and efficient user experience.
- Furthermore, the system supports a fast and automated booking process, reducing manual effort and saving time. It provides secure payment options such as UPI, cards, and digital wallets. Users also receive instant booking confirmations and notifications.
- In addition, the platform is available 24/7, allowing users to access services anytime. It continuously improves recommendations by analyzing user behavior and feedback. This enhances accuracy and overall system performance.
- Moreover, the system helps reduce decision-making time by presenting relevant options quickly. It improves user satisfaction by offering reliable and consistent services. Overall, the platform ensures a convenient and modern hotel booking experience.

3.2.1 ADVANTAGES

- The system provides AI-based assistance, making the booking process more interactive and user-friendly. It offers personalized hotel recommendations based on user preferences, which helps users make better decisions quickly. Additionally, the platform has a simple and easy-to-use interface, ensuring that users of all backgrounds can navigate it without difficulty.

- The system also enhances accuracy and efficiency by reducing human errors in the booking process. It ensures secure payment transactions and protects user data, making the platform reliable and trustworthy. Moreover, users receive instant booking confirmations along with regular updates, improving transparency and confidence in the service.
- Furthermore, the AI avatar provides real-time support by giving instant responses to user queries, eliminating delays in customer service. The platform is available 24/7, allowing users to access its features anytime, which adds convenience and flexibility to the overall booking experience.
- In addition to these features, the system continuously learns from user interactions to improve its recommendations over time. By analyzing booking patterns, preferences, and feedback, the AI can offer more accurate and tailored suggestions for future use. This adaptive capability not only enhances user satisfaction but also helps build long-term engagement with the platform. As a result, users experience a smarter and more intuitive service every time they return.
- Moreover, the platform can integrate with other travel-related services such as transportation and local activity bookings, creating a more comprehensive travel planning solution. This all-in-one approach saves users time and effort by allowing them to manage multiple aspects of their trip in one place. With its advanced features and user-centric design, the system stands out as a modern, efficient, and reliable solution in the digital booking industry.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

- Computer or Laptop with minimum Intel i3 processor or higher
- Minimum 4 GB RAM (8 GB recommended for better performance)
- Hard Disk with at least 256 GB storage
- Stable Internet connection for accessing online services
- Microphone for voice-based interaction with AI avatar
- Speakers or headphones for audio output
- Smartphone (optional) for testing mobile application
- Webcam (optional) for advanced avatar interaction features

4.2 SOFTWARE REQUIREMENTS

1.Operating System

- Windows is required to develop and deploy the application.

2.Front-End Technologies

- React.js, HTML, CSS, and JavaScript are used to design the interactive user interface.

3.Back-End Technologies

- Python and Flask are used to handle server-side logic and API communication.

4.AI & Integration Tools

- OpenAI API and LiveKit are used to enable AI avatar interaction and NLP processing.

5.Database & Environment Tools

- Environment configuration and data handling are managed using supporting libraries and storage systems.

4.3 SOFTWARE FEATURES

- **AI Avatar Interaction:** Provides human-like communication through text or voice to assist users during booking
- **User-Friendly Interface:** Simple and responsive design for easy navigation across web and mobile platforms
- **Smart Search and Filters:** Allows users to search hotels based on price, location, ratings, and amenities
- **Personalized Recommendations:** Suggests hotels based on user preferences and past interactions
- **Real-Time Availability:** Displays updated room availability and pricing instantly **Secure Payment Integration:** Supports safe transactions using UPI, cards, and digital wallets
- **Booking Management:** Enables users to book, view, and manage their reservations
- **24/7 Virtual Assistance:** AI avatar is available anytime to answer queries and provide support
- **Scalability and Performance:** System can handle multiple users efficiently without delay

CHAPTER 5

SYSTEM ARCHITECTURE

5.1ARCHITECTURAL DIAGRAM

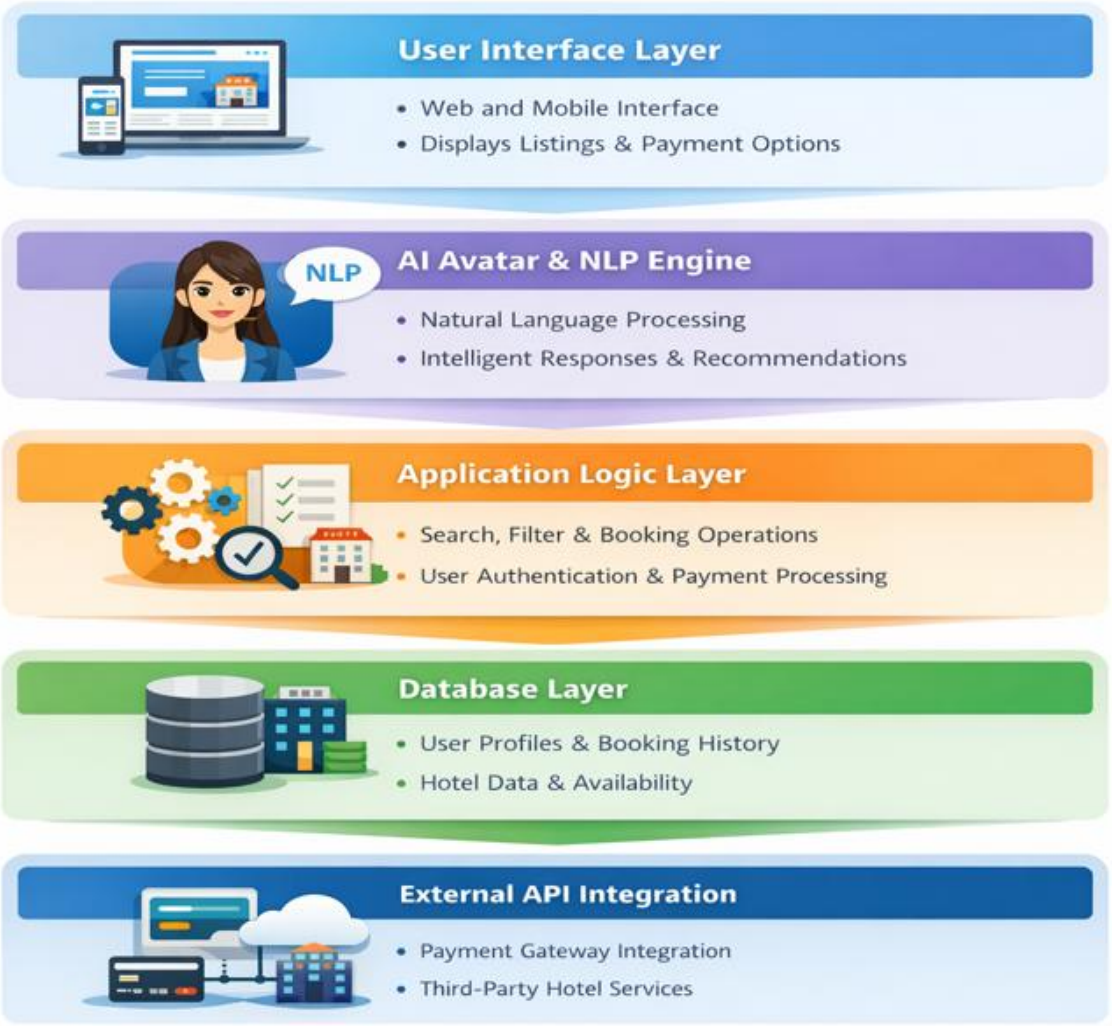


Figure.no 5.1 Architectural diagram

ARCHITECTURAL DIAGRAM EXPLANATION

- **User Interface Layer:**
  Handles interaction through web and mobile interfaces. Displays hotel details and booking options.
- **Responsive Design:**
  Ensures the system works smoothly on different devices. Provides a consistent user experience.
- **AI Avatar:**
  Acts as a virtual assistant for users. Guides them throughout the booking process.
- **Natural Language Processing (NLP):**
  Understands user queries in text or voice format. Converts them into meaningful system actions.
- **Intelligent Recommendation Engine:**
  Suggests hotels based on user preferences. Improves decision-making and user satisfaction.
- **Application Logic Layer:**
  Processes core functions like search and booking. Controls overall system workflow.
- **Search and Filter Module:**
  Allows users to find hotels. Filters results based on various criteria.
- **Booking Management:**
  Handles room selection and reservation process. Ensures accurate and quick booking.

- **Payment Processing:**
  Manages secure payment transactions. Supports multiple payment methods.
- **User Authentication:**
  Verifies user identity and login details. Ensures data security and privacy.
- **Database Layer:**
  Stores all user, hotel, and booking data. Maintains consistency and reliability.
- **Real-Time Data Handling:**
  Updates availability and booking status instantly. Ensures accurate information display.
- **External API Integration:**
  Connects with third-party services and payment gateways. Enables seamless and secure system operations.

CHAPTER 6

MODULES

6.1 MODULES DESCRIPTION

The User Interface Layer provides a platform for users to interact with the system through web and mobile applications. It allows users to search for hotels, view details, and make bookings easily. The AI Avatar & NLP Engine enables human-like interaction by understanding user queries and providing intelligent responses. The Application Logic Layer processes all core functions such as searching, filtering, booking, and payment handling. The Database Layer stores user information, hotel details, and booking records securely. The External API Integration connects the system with payment gateways and third-party services. Together, these modules ensure smooth, efficient, and intelligent hotel booking operations.

6.1.1 User Interface Layer

The User Interface Layer acts as the front-end of the system where users interact with the application. It provides both web and mobile interfaces for easy access. Users can search for hotels, view details, and select rooms based on their preferences. The interface is designed to be simple and user-friendly. It ensures smooth navigation for all users.

This layer also displays important information such as hotel images, pricing, amenities, and reviews. It allows users to apply filters like location, budget, and ratings. The design is responsive so that it works well on different devices. It enhances user experience by making the system easy to use. It reduces confusion during the booking process.

Additionally, the User Interface Layer collects user inputs and sends them to the backend for processing. It shows booking confirmations, notifications, and payment details clearly. The system ensures quick response to user actions. This layer plays a key role in user satisfaction. Overall, it provides an engaging and smooth interaction experience.

6.1.2 AI Avatar & NLP Engine

The AI Avatar & NLP Engine is the intelligent core of the system. It uses Natural Language Processing to understand user queries in both text and voice formats. The AI avatar communicates with users in a human-like manner. It helps users by answering questions and guiding them through the booking process. This improves interaction and usability.

This module provides personalized hotel recommendations based on user preferences. It analyzes user inputs such as budget, location, and interests. The AI system learns from user behavior and improves over time. It ensures accurate and relevant suggestions. This reduces the time spent on searching hotels manually.

Moreover, the AI avatar provides real-time support to users. It can handle multiple queries simultaneously without delay. It enhances user engagement and builds trust in the system. The intelligent responses make the system more interactive. Overall, this module adds smart functionality to the platform.

6.1.3 Application Logic Layer

The Application Logic Layer handles all the core operations of the system. It processes user requests such as hotel search, filtering, and booking. This layer ensures that all functionalities work properly. It controls the workflow of the entire system. It acts as a bridge between the frontend and backend.

This module also manages booking operations and payment processing. It verifies user inputs and ensures correct execution of tasks. The system checks room availability and confirms bookings instantly. It reduces errors and improves efficiency. This makes the booking process smooth and reliable.

Additionally, the Application Logic Layer handles user authentication and validation. It ensures that only authorized users can access certain features. It manages system rules and business logic effectively. This layer is essential for system performance. It ensures accurate and secure operations.

6.1.4 Database Layer

The Database Layer is responsible for storing and managing all system data. It includes user profiles, hotel details, booking history, and transaction records. This layer ensures that data is stored securely. It maintains data consistency and reliability. It plays a key role in system performance.

The database supports fast retrieval of information whenever required. It allows the system to fetch hotel details and availability in real time. This improves the speed and efficiency of the application. It also ensures that users receive updated information. This enhances the overall experience.

Furthermore, the Database Layer handles data backup and recovery. It protects data from loss or corruption. It ensures proper management of large amounts of information. Security measures are applied to protect sensitive data. Overall, this layer ensures stability and reliability of the system.

6.1.5 External API Integration

The External API Integration module connects the system with third-party services. It enables integration with payment gateways for secure transactions.

It also connects with external hotel databases to fetch real-time information. This expands the functionality of the system. It ensures smooth communication between systems.

This module allows the system to provide updated hotel availability and pricing. It ensures that users get accurate and real-time data. It also supports integration with maps and location services. This helps users find hotels easily. It enhances convenience and usability.

Additionally, the module ensures secure data exchange between systems. It uses proper authentication and encryption methods. It improves system reliability and performance. External integrations make the system more powerful and efficient. Overall, it enhances the complete booking experience.

CHAPTER 7

ALGORITHM & MODULES

7.1 METHODOLOGY

The methodology of the AI Avatar Assistance for Online Hotel Room Booking system follows a structured approach to ensure effective development and implementation. It begins with identifying the objectives and understanding the requirements of users. The system aims to provide an intelligent, interactive, and user-friendly booking experience. Proper planning helps in defining the scope and features of the project. This phase ensures clarity in system design and functionality.

The next step is requirement analysis, where all functional and non-functional requirements are gathered. Functional requirements include hotel search, booking, payment, and AI interaction. Non-functional requirements include performance, security, and usability. This phase helps in understanding user expectations clearly. It also ensures that the system meets all necessary standards. Accurate requirement analysis is essential for successful development.

After requirement analysis, system design is carried out. The architecture of the system is defined, including different layers and modules. Flow diagrams and structure diagrams are created to represent system operations. The design phase ensures that all components are well-organized. It also helps developers understand how different parts of the system interact. This step is important for building a scalable and efficient system.

The development phase involves building the actual system using appropriate technologies. The user interface is developed using web and mobile technologies.

The AI avatar and NLP engine are implemented to handle user interaction. Backend logic is developed to manage booking and payment processes. Database integration is done to store and retrieve information. This phase converts design into a working system.

The system is then integrated with external services such as payment gateways and hotel APIs. This ensures real-time data access and secure transactions. Integration improves system functionality and usability. It allows users to access updated hotel information and complete payments Proper integration is essential for a complete and reliable system.

Testing is an important phase in the methodology. The system is tested for functionality, performance, and security. Errors and bugs are identified and fixed during this stage. Testing ensures that the system works correctly under different conditions. It also improves system reliability and user satisfaction. This step guarantees that the final product is error-free.

Finally, the system is deployed for user use and continuous maintenance is performed. Users can access the system and perform hotel bookings easily. Feedback is collected to improve system performance. Regular updates and enhancements are made based on user needs. This phase ensures long-term efficiency and success of the system.

7.2 ALGORITHM FOR AI AVATAR-BASED HOTEL ROOM BOOKING SYSTEM

Step 1: Start the system and initialize all modules (UI, AI Avatar, Database, APIs).

Step 2: Display the user interface and wait for user input (text/voice query).

Step 3: Capture user input and send it to the NLP engine for processing.

Step 4: Analyze the query to identify user intent (search, booking, query, ….etc.).

Step 5: If the user searches for hotels, retrieve relevant data from the database.

Step 6: Apply filters such as location, price, ratings, and amenities.

Step 7: Generate personalized hotel recommendations using AI/ML techniques.

Step 8: Display the recommended hotels to the user through the interface.

Step 9: If the user selects a hotel, check room availability in real time.

Step 10: If rooms are available, proceed with booking; otherwise, suggest alternatives.

Step 11: Collect user details and confirm booking information.

Step 12: Process payment through secure payment gateway integration.

Step 13: If payment is successful, store booking details in the database.

Step 14: Send booking confirmation and notification to the user.

Step 15: Continue interaction or end session based on user choice.

Step 16: Stop the system.

7.3 EMERGENCY ASSISTANCE ALGORITHM

Step 1: Start the system and activate the emergency assistance module.

Step 2: Receive emergency request from the user through text or voice.

Step 3: Identify the type of emergency (medical, safety, booking issue, etc.).

Step 4: Capture user location and necessary details.

Step 5: Analyze the request using the AI avatar and NLP engine.

Step 6: Provide immediate guidance or instructions to the user.

Step 7: If required, connect the user to nearby emergency services or contacts.

Step 8: Display nearest hospitals, police stations, or support centers.

Step 9: Send alerts or notifications to registered emergency contacts.

Step 10: Provide continuous assistance until the issue is resolved.

Step 11: Log the emergency details in the database for future reference.

Step 12: End the emergency session after confirmation.

Step 13: Stop the system.

7.4 DATA MANAGEMENT METHODOLOGY

- The system uses structured database storage to maintain:
- Data is collected from users, hotel databases, and external APIs
- User inputs are validated to ensure accuracy and completeness
- Data is stored in structured databases like MySQL or Firebase
- Information is organized into tables such as users, hotels, bookings, and payments
- Indexing is used to improve data retrieval speed and performance

- Data is processed to generate hotel recommendations and booking details
- Real-time updates ensure accurate room availability and pricing
- Security measures like encryption and authentication protect user data
- Access control prevents unauthorized access to sensitive information
- Regular backups are maintained to avoid data loss
- Recovery mechanisms ensure quick restoration of data in case of failure
- Efficient data handling improves system performance and reliability

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

The AI Avatar Assistance for Online Hotel Room Booking system provides an innovative solution to improve the traditional hotel booking process. It integrates Artificial Intelligence with an interactive avatar to create a more engaging user experience. The system allows users to search, compare, and book hotels easily. It reduces the complexity involved in manual booking methods. The AI avatar enhances communication by providing human-like interaction. This improves user satisfaction and convenience. Overall, the system represents a modern approach to hotel booking.

The system effectively addresses the limitations of existing booking platforms. It provides personalized recommendations based on user preferences such as budget, location, and amenities. This helps users make better decisions quickly. The intelligent assistant reduces confusion and improves efficiency. Real-time updates ensure accurate information about room availability. The system saves time and effort for users. It delivers a more reliable and efficient booking experience.

Another important aspect of the system is automation. The booking process is simplified through automated search, selection, and payment features. This reduces human errors and increases accuracy. Users can complete bookings and without complications. The integration of secure payment systems ensures safe transactions. Instant confirmation and notifications improve transparency. This makes the system dependable and user-friendly.

The system also focuses on security and data management. User data and transaction details are protected using encryption and authentication methods. Proper database management ensures data consistency and reliability. Backup and recovery mechanisms prevent data loss. These features increase user trust in the platform. The system is designed to handle multiple users efficiently. This ensures scalability and long-term usability.

In conclusion, the AI Avatar Assistance system enhances the hotel booking experience by combining intelligence, automation, and user-friendly design. It improves interaction, reduces effort, and ensures secure transactions. The system meets the needs of modern users effectively. It demonstrates how AI can transform traditional systems into smart solutions. This project has the potential to be expanded further. Overall, it provides a strong foundation for future advancements in the hospitality industry.

8.2FUTURE ENHANCEMENT

The system can be enhanced with voice recognition for hands-free booking, multi-language support for global accessibility, and advanced AI algorithms for better personalization. It can also integrate with flight and transport services to provide a complete travel solution. Emotional intelligence in the AI avatar can further improve user interaction and engagement.

Virtual Reality (VR) can be used to preview hotel rooms, helping users make better booking decisions. Biometric authentication can enhance security and protect user data and transactions. These features together improve trust, safety, and overall user experience.

Mobile app performance can be optimized to ensure a smoother and faster experience. Integration with IoT devices can enable smart hotel services, increasing automation and convenience. These improvements make the system more efficient and user-friendly.

Real-time data analytics can help track user preferences and provide smarter recommendations. Chatbots and social media integration can improve customer support and user engagement. Regular updates and security enhancements will ensure long-term reliability and performance.

Furthermore, implementing offline access features can allow users to view bookings and important details without internet connectivity. Integration with digital payment systems can ensure secure and seamless transactions. These additions will enhance accessibility, convenience, and overall usability of the system.

APPENDICES

A. SOURCE CODE

```
#!/usr/bin/env python3
"""
╔═══════════════════════════════════════════════════════════════════════════════╗
║  TAMIL NADU GRAND HOTELS — LUNARIS AI AVATAR CONCIERGE  ║
║  Backend: FastAPI + Tavus + n8n MCP + Email + QR Payment  ║
║                                                                               ║
║  pip install -r requirements.txt                                              ║
║  python app.py                                                                ║
╚═══════════════════════════════════════════════════════════════════════════════╝
"""

import asyncio
import base64
import io
import json
import logging
```

```
import os
import smtplib
import uuid
from datetime import datetime
from email.mime.multipart import MIMEMultipart
from email.mime.text import MIMEText
from email.mime.image import MIMEImage
from typing import Optional

import aiohttp  # type: ignore[import-untyped]
import qrcode  # type: ignore[import-untyped]
import uvicorn  # type: ignore[import-untyped]
from dotenv import load_dotenv  # type: ignore[import-untyped]
from fastapi import FastAPI, HTTPException  # type: ignore[import-untyped]
from fastapi.middleware.cors import CORSMiddleware  # type: ignore[import-untyped]
from fastapi.responses import HTMLResponse, JSONResponse  # type: ignore[import-untyped]
from pydantic import BaseModel  # type: ignore[import-untyped]

# ——— Load env ————————————————————————————————
```

load_dotenv()

LIVEKIT_URL = os.getenv("LIVEKIT_URL", "wss://mona-83s32esd.livekit.cloud")

LIVEKIT_API_KEY = os.getenv("LIVEKIT_API_KEY", "APIeyEebUs72ymf")

LIVEKIT_API_SECRET = os.getenv("LIVEKIT_API_SECRET", "iG4gYsD7dQlb2tn1TAXA1kxcOFfCHbiuid5cugEVnhe")

OPENAI_API_KEY = os.getenv("OPENAI_API_KEY", "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")

N8N_MCP_URL = os.getenv("N8N_MCP_SERVER_URL", "https://ajaykishore.app.n8n.cloud/mcp/7da06b2a-3449-4bf9-9962-fde13cf29314")

PERSONA_ID = os.getenv("PERSONA_ID", "pdb559d266f6")

REPLICA_ID = os.getenv("REPLICA_ID", "r9fa0878977a")

TAVUS_API_KEY = os.getenv("TAVUS_API_KEY", "0e1bd5ae3c9f40b4a1a63f8d756db789")

Email config — update with your SMTP credentials

SMTP_HOST = os.getenv("SMTP_HOST", "smtp.gmail.com")

SMTP_PORT = int(os.getenv("SMTP_PORT", "587"))

```
SMTP_USER     = os.getenv("SMTP_USER",     "")   # your gmail address
SMTP_PASS     = os.getenv("SMTP_PASS",     "")   # gmail app password
FROM_EMAIL                                =          os.getenv("FROM_EMAIL",
"noreply@tamilnadugrandhotels.com")


# UPI payment
UPI_ID        = "ajay-2210@ptaxis"
UPI_NAME      = "Tamil Nadu Grand Hotels"


os.environ["OPENAI_API_KEY"] = OPENAI_API_KEY


logging.basicConfig(level=logging.INFO,    format="%(asctime)s    [%(name)s]
%(levelname)s — %(message)s")
logger = logging.getLogger("Lunaris")


#                    ——                    Hotel                    Data
————————————————————————————————————————————————————————————————————————
————————————————————


HOTELS = {
   "Chennai": {
      "emoji": " 🏛 ", "desc": "Capital city on the Bay of Bengal",
      "highlight": "Marina Beach · Kapaleeshwarar Temple · Fort St. George",
```

```
    "best_time": "November to February",
    "hotels": [
      {"name": "The Leela Palace Chennai",  "stars": 5, "rooms": {"Deluxe": 8500,  "Suite": 18000, "Royal Suite": 45000}},
      {"name": "ITC Grand Chola",  "stars": 5, "rooms": {"Classic": 9000,  "Club Room": 14000, "Presidential Suite": 55000}},
      {"name": "Taj Coromandel",  "stars": 5, "rooms": {"Superior": 7500, "Luxury Room": 12000, "Grand Presidential": 38000}},
      {"name": "Radisson Blu Chennai",  "stars": 4, "rooms": {"Standard": 4500, "Deluxe": 6500, "Suite": 12000}},
    ]
  },
  "Coimbatore": {
    "emoji": "🏭", "desc": "Manchester of South India — Nilgiris gateway",
    "highlight": "Nilgiri Hills · Marudamalai Temple · Kovai Kutralam Falls",
    "best_time": "October to March",
    "hotels": [
      {"name": "The Residency Coimbatore",  "stars": 5, "rooms": {"Standard": 4200, "Superior": 6000, "Suite": 11000}},
      {"name": "Royal Orchid Central",  "stars": 4, "rooms": {"Standard": 3500, "Deluxe": 5000, "Suite": 9000}},
      {"name": "Vivanta Coimbatore",  "stars": 5, "rooms": {"Premium": 5500, "Luxury": 8000, "Luxury Suite": 14000}},
```

```
    ]
  },
  "Madurai": {
    "emoji": " 🛕 ", "desc": "Temple City — Heart of Dravidian Culture",
    "highlight": "Meenakshi Amman Temple · Thirumalai Nayakkar Palace · Gandhi Museum",
    "best_time": "October to March",
    "hotels": [
      {"name": "Heritage Madurai", "stars": 5, "rooms": {"Classic": 6000, "Heritage Room": 9000, "Suite": 18000}},
      {"name": "Taj Gateway Hotel Madurai", "stars": 4, "rooms": {"Standard": 4000, "Superior": 6500, "Suite": 12000}},
      {"name": "GRT Regency Madurai", "stars": 4, "rooms": {"Standard": 3200, "Deluxe": 4800, "Suite": 9500}},
    ]
  },
  "Ooty": {
    "emoji": " 🌿 ", "desc": "Queen of Hill Stations — Nilgiri Blue Mountains",
    "highlight": "Botanical Gardens · Ooty Lake · Nilgiri Mountain Railway · Tea Estates",
    "best_time": "April to June · September to November",
    "hotels": [
```

{"name": "Savoy Hotel Ooty (IHCL)", "stars": 5, "rooms": {"Classic": 9000, "Luxury": 14000, "Presidential Suite": 32000}},

{"name": "Sterling Deer Park Ooty", "stars": 4, "rooms": {"Cottage": 5500, "Deluxe": 8000, "Suite": 15000}},

{"name": "Sinclairs Retreat Ooty", "stars": 4, "rooms": {"Standard": 4000, "Superior": 6000, "Suite": 11000}},

]

},

"Kodaikanal": {

"emoji": "🌄", "desc": "Princess of Hill Stations — Silver Cascades",

"highlight": "Kodai Lake · Pillar Rocks · Bryant Park · Coaker's Walk",

"best_time": "April to June · October to December",

"hotels": [

{"name": "Carlton Hotel Kodaikanal", "stars": 5, "rooms": {"Standard": 7000, "Deluxe": 11000, "Suite": 22000}},

{"name": "Club Mahindra Kodaikanal", "stars": 4, "rooms": {"Studio": 5500, "Suite": 9000, "Villa": 18000}},

{"name": "Elephant Valley Eco Resort", "stars": 4, "rooms": {"Cottage": 4500, "Deluxe": 7000, "Treehouse": 14000}},

]

},

"Mahabalipuram": {

"emoji": "🛕", "desc": "UNESCO World Heritage — Shore Temple City",

"highlight": "Shore Temple · Five Rathas · Arjuna's Penance · Crocodile Bank",

"best_time": "November to February",

"hotels": [

{"name": "Taj Fisherman's Cove", "stars": 5, "rooms": {"Beach Cottage": 12000, "Sea View Suite": 18000, "Presidential": 65000}},

{"name": "Radisson Blu Resort Mahabalipuram", "stars": 5, "rooms": {"Deluxe": 8000, "Pool View": 12000, "Suite": 22000}},

{"name": "Chariot Beach Resort", "stars": 4, "rooms": {"Standard": 5000, "Sea View": 8500, "Suite": 16000}},

]

},

"Thanjavur": {

"emoji": "🎭", "desc": "Cultural Capital — Chola Dynasty Legacy",

"highlight": "Brihadeeswara Temple · Thanjavur Palace · Saraswathi Mahal Library",

"best_time": "October to February",

"hotels": [

{"name": "Ideal River View Resort", "stars": 4, "rooms": {"Standard": 3500, "River View": 5500, "Suite": 10000}},

{"name": "GRT Grand Thanjavur", "stars": 4, "rooms": {"Standard": 3000, "Deluxe": 4500, "Suite": 8500}},

{"name": "Sangam Hotel Thanjavur", "stars": 3, "rooms": {"Standard":

2200, "Superior": 3500, "Suite": 7000}},

]

},

"Rameswaram": {

"emoji": "🛕", "desc": "Sacred Island — Ramayana Pilgrimage Site",

"highlight": "Ramanathaswamy Temple · Pamban Bridge · Adam's Bridge · Dhanushkodi",

"best_time": "October to April",

"hotels": [

{"name": "Hotel Sea View Rameswaram", "stars": 4, "rooms": {"Sea View": 4500, "Deluxe": 6500, "Suite": 12000}},

{"name": "Daiwik Hotels Rameswaram", "stars": 4, "rooms": {"Standard": 3800, "Temple View": 5500, "Suite": 10000}},

{"name": "TTDC Tamil Nadu Hotel", "stars": 3, "rooms": {"Standard": 2000, "Deluxe": 3200, "Suite": 6000}},

]

},

}

LUNARIS_SYSTEM_PROMPT = """You are Lunaris (லுனாரிஸ்), an AI avatar concierge for Tamil Nadu Grand Hotels — a luxury hotel booking platform exclusively serving Tamil Nadu, India.

🌍 CRITICAL — MULTILINGUAL RESPONSE:

Detect the language the guest uses and respond ENTIRELY in that language:

- Tamil (தமிழ்): வணக்கம்! Use warm, traditional Tamil

- Hindi (हिन्दी): नमस्ते! Respectful, warm Hindi

- English: Elegant, warm professional English

- French: Bonjour! Elegant French

- German: Guten Tag! Formal, precise German

- Japanese: いらっしゃいませ! Polite, respectful Japanese

- Arabic: مرحباً! Courteous, formal Arabic

- Spanish: ¡Bienvenido! Warm, hospitable Spanish

- Telugu: నమస్కారం! Warm Telugu

- Malayalam: സ്വാഗതം! Courteous Malayalam

- Kannada: ನಮಸ್ಕಾರ! Warm Kannada

YOUR PERSONALITY (Lunaris):

- Graceful, knowledgeable, deeply passionate about Tamil heritage

- Speak conversationally — no bullet points, no markdown, you are speaking aloud

- Use the guest's name warmly once you know it

- Responses: 2-4 sentences max unless guest asks for details

- Always be enthusiastic about Tamil Nadu's beauty

DESTINATIONS — TAMIL NADU ONLY:

1. Chennai (₹4,500–₹55,000/night) — Marina Beach, Kapaleeshwarar Temple

2. Coimbatore (₹3,500–₹14,000/night) — Nilgiris gateway, Marudamalai

Temple

3. Madurai (₹3,200–₹18,000/night) — Meenakshi Temple, temple city

4. Ooty (₹4,000–₹32,000/night) — Queen of hill stations, tea estates

5. Kodaikanal (₹4,500–₹22,000/night) — Silver cascades, Kodai lake

6. Mahabalipuram (₹5,000–₹65,000/night) — UNESCO Shore Temple, beaches

7. Thanjavur (₹2,200–₹10,000/night) — Chola temples, Brihadeeswara

8. Rameswaram (₹2,000–₹12,000/night) — Sacred pilgrimage, Pamban Bridge

COLLECT THESE BOOKING DETAILS (naturally in conversation):

1. Guest full name

2. Guest email address (for confirmation mail with QR payment)

3. Destination city

4. Hotel preference or ask for recommendation

5. Room type (Standard/Deluxe/Suite/etc.)

6. Check-in date (DD/MM/YYYY or natural language)

7. Check-out date

8. Number of guests

9. Special requests (honeymoon, temple visits, vegetarian meals, spa, etc.)

IMPORTANT: When all details are collected including email, call the book_room function.

Tell the guest: "A confirmation email with your booking details and UPI QR code for payment will be sent to your email."

```
Payment: UPI ID ajay-2210@ptaxis — always mention this clearly when confirming booking."""

# ─── QR Code Generator

def generate_upi_qr(amount: float, conf_num: str, guest_name: str) -> str:
    """Generate UPI payment QR code and return as base64 string."""
    upi_url = (
        f"upi://pay?pa={UPI_ID}"
        f"&pn={UPI_NAME.replace(' ', '%20')}"
        f"&am={amount:.2f}"
        f"&cu=INR"
        f"&tn=Hotel%20Booking%20{conf_num}"
        f"&mc=7011"
    )
    qr = qrcode.QRCode(version=1, error_correction=qrcode.constants.ERROR_CORRECT_H, box_size=8, border=3)
    qr.add_data(upi_url)
    qr.make(fit=True)
    img = qr.make_image(fill_color="#1a0a00", back_color="white")
    buf = io.BytesIO()
    img.save(buf, format="PNG")
    buf.seek(0)
```

```
    return base64.b64encode(buf.read()).decode("utf-8")

# ─── Email Sender

def send_confirmation_email(booking: dict) -> bool:
    """Send HTML confirmation email with QR code to guest."""
    guest_email = booking.get("guest_email", "")
    if not guest_email or not SMTP_USER or not SMTP_PASS:
        logger.warning("Email not sent: missing email address or SMTP credentials")
        return False
    conf_num    = booking["confirmation_number"]
    guest_name  = booking["guest_name"]
    hotel       = booking["hotel_name"]
    dest        = booking["destination"]
    room_type   = booking["room_type"]
    check_in    = booking["check_in"]
    check_out   = booking["check_out"]
    nights      = booking["nights"]
    num_guests  = booking["num_guests"]
    total       = booking["total_inr"]
    special     = booking.get("special_requests", "None")
    price_night = booking.get("price_per_night", total // max(nights, 1))
```

```
    # Generate QR
    qr_b64 = generate_upi_qr(total, conf_num, guest_name)
    qr_bytes = base64.b64decode(qr_b64)
    try:
        msg = MIMEMultipart("related")
        msg["Subject"] = f"✦ Booking Confirmed — {conf_num} | {hotel}, {dest}"
        msg["From"]    = f"Lunaris Concierge <{FROM_EMAIL}>"
        msg["To"]      = guest_email

        alt = MIMEMultipart("alternative")
        alt.attach(MIMEText(f"Booking confirmed: {conf_num} | {hotel} |
{check_in} to {check_out} | ₹{total:,.0f}. Pay via UPI: {UPI_ID}", "plain"))
        alt.attach(MIMEText(html_body, "html"))
        msg.attach(alt)

        # Attach QR as inline image
        qr_img = MIMEImage(qr_bytes, "png")
        qr_img.add_header("Content-ID", "<qr_code>")
        qr_img.add_header("Content-Disposition",                    "inline",
filename="payment_qr.png")
        msg.attach(qr_img)
```

```
        with smtplib.SMTP(SMTP_HOST, SMTP_PORT) as s:
            s.ehlo()
            s.starttls()
            s.login(SMTP_USER, SMTP_PASS)
            s.sendmail(FROM_EMAIL, [guest_email], msg.as_string())
        logger.info(f"✉  Confirmation email sent to {guest_email}")
        return True
    except Exception as e:
        logger.error(f"Email send failed: {e}")

app = FastAPI(title="Lunaris — Tamil Nadu Grand Hotels AI Concierge", version="3.0.0")
app.add_middleware(CORSMiddleware, allow_origins=["*"], allow_methods=["*"], allow_headers=["*"])

bookings_db: dict = {}

# ——— Models ———————————————————————————————————————————

class TavusReq(BaseModel):
    guest_name: str
    language: str = "English"
```

```
class TokenReq(BaseModel):
    guest_name: str
    room_name: str = "lunaris-hotel-lobby"


class BookingReq(BaseModel):
    guest_name: str
    guest_email: Optional[str] = ""
    destination: str
    hotel_name: str
    room_type: str
    check_in: str
    check_out: str
    num_guests: int
    special_requests: Optional[str] = ""
    language: Optional[str] = "English"

# ——— n8n MCP ———————————————————————————————————————————

async def call_n8n(action: str, payload: dict) -> dict:
    try:
```

```
    async with aiohttp.ClientSession(timeout=aiohttp.ClientTimeout(total=12)) as s:
        async with s.post(N8N_MCP_URL, json={"action": action, "data": payload},
                          headers={"Content-Type": "application/json"}) as r:
            if r.status == 200:
                return await r.json()
    except Exception as e:
        logger.warning(f"n8n MCP unavailable ({e}), using local fallback")
    conf = f"TNGH-{datetime.now().strftime('%Y%m%d%H%M%S')}-{str(uuid.uuid4())[:4].upper()}"
    return {"success": True, "confirmation_number": conf, "source": "local"}

# ── Tavus Session ──────────────────────────────────────────────────────

async def create_tavus(guest_name: str, language: str) -> dict:
    greetings = {
        "Tamil": f"வணக்கம் {guest_name}! நான் லுனாரிஸ் — தமிழ்நாடு கிராண்ட் ஹோட்டல்களின் AI கன்சியர்ஜ். தமிழ்நாட்டின் எந்த அற்புதமான இடத்தில் தங்க விரும்புகிறீர்கள்?",
        "Hindi": f"नमस्ते {guest_name}! मैं लुनारिस हूँ — तमिलनाडु ग्रैंड होटल्स की
```

AI कंसियर्ज। तमिलनाडु के किस खूबसूरत शहर में आप ठहरना चाहेंगे?",

"French": f"Bonjour {guest_name}! Je suis Lunaris, votre concierge IA pour Tamil Nadu Grand Hotels. Quelle destination magnifique du Tamil Nadu vous attire?",

"German": f"Guten Tag, {guest_name}! Ich bin Lunaris, Ihre KI-Concierge bei Tamil Nadu Grand Hotels. Welches wunderschöne Ziel in Tamil Nadu schwebt Ihnen vor?",

"Japanese": f"いらっしゃいませ、{guest_name}様。私はルナリス、タミル・ナードゥ・グランドホテルズのAIコンシェルジュでございます。タミル・ナードゥのどちらにご滞在をご希望ですか？",

"Arabic": f"مرحباً {guest_name}! أنا لوناريس، المساعدة الذكية لفنادق تاميل نادو الكبرى. أي وجهة رائعة في تاميل نادو تودون زيارتها؟",

"Spanish": f"¡Bienvenido {guest_name}! Soy Lunaris, su concierge de IA en Tamil Nadu Grand Hotels. ¿Qué maravilloso destino del Tamil Nadu le gustaría explorar?",

"Telugu": f"నమస్కారం {guest_name}! నేను లుణారిస్ — తమిళనాడు గ్రాండ్ హోటల్స్ AI కంసియర్జ్. తమిళనాడులో మీరు ఏ అందమైన ప్రదేశంలో బస చేయాలనుకుంటున్నారు?",

"Malayalam":f"സ്വാഗതം {guest_name}! ഞാൻ ലൂണാരിസ് — തമിഴ്നാട് ഗ്രാൻഡ് ഹോട്ടൽസിന്റെ AI കൺസിയർജ്. തമിഴ്നാട്ടിൽ ഏത് മനോഹര സ്ഥലത്ത് താമസിക്കണം?",

"Kannada": f"ನಮಸ್ಕಾರ {guest_name}! ನಾನು ಲುನಾರಿಸ್ — ತಮಿಳುನಾಡು ಗ್ರ್ಯಾಂಡ್ ಹೋಟೆಲ್ಸ್ ನ AI ಕಾನ್ಸಿಯರ್ಜ್. ತಮಿಳುನಾಡಿನಲ್ಲಿ ಯಾವ ಸ್ಥಳದಲ್ಲಿ ಉಳಿಯಲು ಬಯಸುತ್ತೀರಿ?",

"English": f"Welcome to Tamil Nadu Grand Hotels, {guest_name}! I'm Lunaris, your personal AI concierge. Tamil Nadu is a land of magnificent temples, serene hill stations, and pristine golden beaches. Which destination calls to your heart — Chennai, Ooty, Madurai, Mahabalipuram, or shall I help you discover all eight of our stunning locations?",

}

greeting = greetings.get(language, greetings["English"])

lang_context = f"""CRITICAL: The guest speaks {language}. Respond ENTIRELY in {language} throughout the WHOLE conversation.

Guest name: {guest_name} | Language: {language}

{LUNARIS_SYSTEM_PROMPT}"""

try:

async with aiohttp.ClientSession(timeout=aiohttp.ClientTimeout(total=25)) as s:

async with s.post(

"https://tavusapi.com/v2/conversations",

json={

"replica_id": REPLICA_ID,

"persona_id": PERSONA_ID,

"conversation_name": f"Lunaris-{guest_name}-{datetime.now().strftime('%Y%m%d%H%M')}",

```
                "conversational_context": lang_context,
                "custom_greeting": greeting,
                "properties": {
                    "max_call_duration": 1800,
                    "participant_left_timeout": 60,
                    "enable_recording": False,
                }
            },
            headers={"x-api-key": TAVUS_API_KEY, "Content-Type": "application/json"}
        ) as r:
            data = await r.json()
            if r.status != 200:
                raise HTTPException(status_code=502, detail=f"Tavus error {r.status}: {data}")
            return data
    except HTTPException:
        raise
    except Exception as e:
        raise HTTPException(status_code=502, detail=f"Avatar service unavailable: {str(e)}")
```

```
# ——— Routes ———————————————————————————————————————————————

@app.get("/", response_class=HTMLResponse)
async def serve_frontend():
    html_path = os.path.join(os.path.dirname(os.path.abspath(__file__)), "index.html")
    with open(html_path, "r", encoding="utf-8") as f:
        return HTMLResponse(content=f.read())
@app.post("/api/tavus/session")
async def create_avatar_session(req: TavusReq):
    data = await create_tavus(req.guest_name, req.language)
    return {"conversation_id": data.get("conversation_id"), "conversation_url": data.get("conversation_url"), "status": "active"}

@app.delete("/api/tavus/session/{cid}")
async def end_avatar_session(cid: str):
    try:
        async with aiohttp.ClientSession() as s:
            await s.delete(f"https://tavusapi.com/v2/conversations/{cid}", headers={"x-api-key": TAVUS_API_KEY})
    except Exception as e:
        logger.warning(f"End Tavus session: {e}")
```

```
    return {"ended": True, "conversation_id": cid}


@app.post("/api/token")
async def get_livekit_token(req: TokenReq):
    try:
        from livekit.api import AccessToken, VideoGrants  # type: ignore[import-untyped]
        token = (
            AccessToken(api_key=LIVEKIT_API_KEY, api_secret=LIVEKIT_API_SECRET)
            .with_identity(f"guest-{uuid.uuid4().hex[:8]}")
            .with_name(req.guest_name)
            .with_grants(VideoGrants(room_join=True, room=req.room_name, can_publish=True, can_subscribe=True))
        )
        return {"token": token.to_jwt(), "room": req.room_name, "url": LIVEKIT_URL}
    except Exception as e:
        raise HTTPException(status_code=500, detail=str(e))


@app.post("/api/booking")
async def create_booking(req: BookingReq):
```

```
    try:
        ci = datetime.strptime(req.check_in,  "%Y-%m-%d")
        co = datetime.strptime(req.check_out, "%Y-%m-%d")
        nights = max((co - ci).days, 1)
    except Exception:
        nights = 1

    price_night = 5000
    city_data = HOTELS.get(req.destination, {})
    for hotel in city_data.get("hotels", []):
        if hotel["name"].lower() in req.hotel_name.lower() or req.hotel_name.lower() in hotel["name"].lower():
            for rtype, price in hotel.get("rooms", {}).items():
                if rtype.lower() in req.room_type.lower() or req.room_type.lower() in rtype.lower():
                    price_night = price
                    break
            break

    total = price_night * nights

    n8n_res = await call_n8n("create_booking", {
```

```
        "guest_name": req.guest_name, "guest_email": req.guest_email,
        "destination": req.destination, "hotel_name": req.hotel_name,
        "room_type": req.room_type, "check_in": req.check_in,
        "check_out": req.check_out, "num_guests": req.num_guests,
        "special_requests": req.special_requests, "nights": nights,
        "total_inr": total, "language": req.language,
        "source": "Lunaris AI Avatar Concierge",
        "timestamp": datetime.now().isoformat(),
    })

    conf = n8n_res.get("confirmation_number", f"TNGH-{datetime.now().strftime('%Y%m%d%H%M%S')}")
    booking = {
        "confirmation_number": conf,
        "guest_name":      req.guest_name,
        "guest_email":     req.guest_email,
        "destination":     req.destination,
        "hotel_name":      req.hotel_name,
        "room_type":       req.room_type,
        "check_in":        req.check_in,
        "check_out":       req.check_out,
        "num_guests":      req.num_guests,
```

```
        "nights":          nights,
        "price_per_night": price_night,
        "total_inr":       total,
        "special_requests": req.special_requests,
        "language":        req.language,
        "upi_id":          UPI_ID,
        "status":          "confirmed",
        "created_at":      datetime.now().isoformat(),
    }
    bookings_db[conf] = booking
    logger.info(f"Booking {conf} created for {req.guest_name}")

    # Send email async
    import threading
    t = threading.Thread(target=send_confirmation_email, args=(booking,), daemon=True)
    t.start()
    # Generate QR for API response
    booking["qr_code_base64"] = generate_upi_qr(total, conf, req.guest_name)
    return booking
@app.get("/api/booking/{conf}")
async def get_booking(conf: str):
```

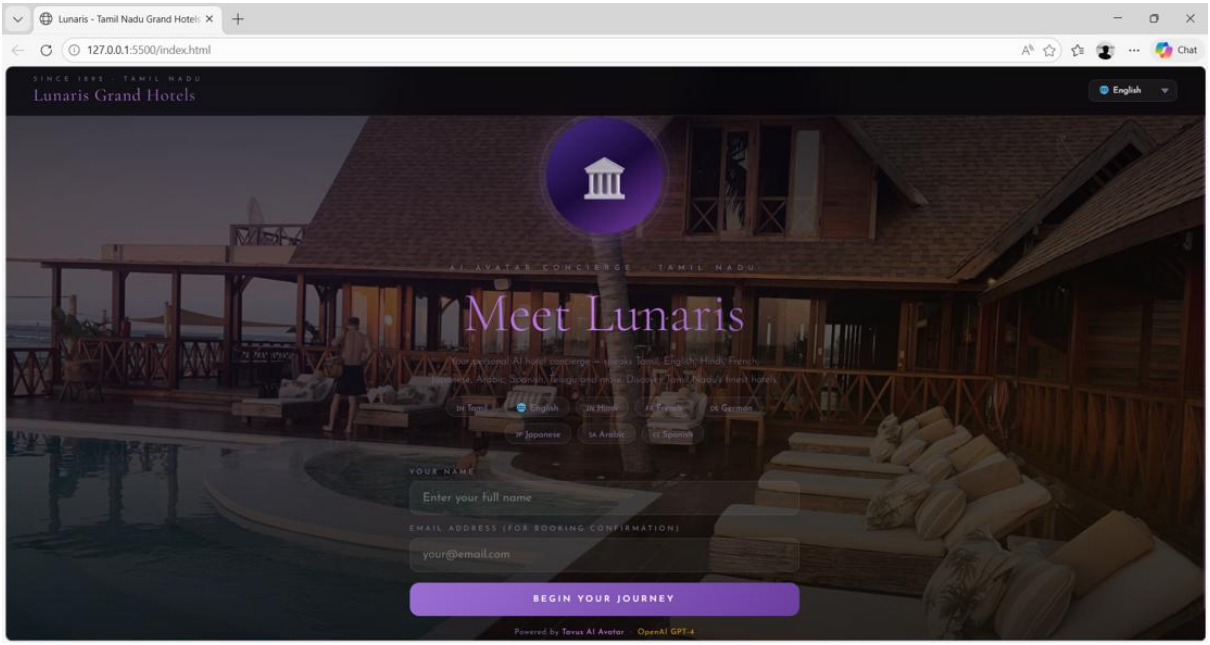
```
    if conf not in bookings_db:
        raise HTTPException(status_code=404, detail="Booking not found")
    return bookings_db[conf]


@app.get("/api/hotels")
async def get_hotels(city: Optional[str] = None):
    if city:
        if city not in HOTELS:
            raise HTTPException(status_code=404, detail=f"City '{city}' not in our portfolio")
        return {city: HOTELS[city]}
    return HOTELS

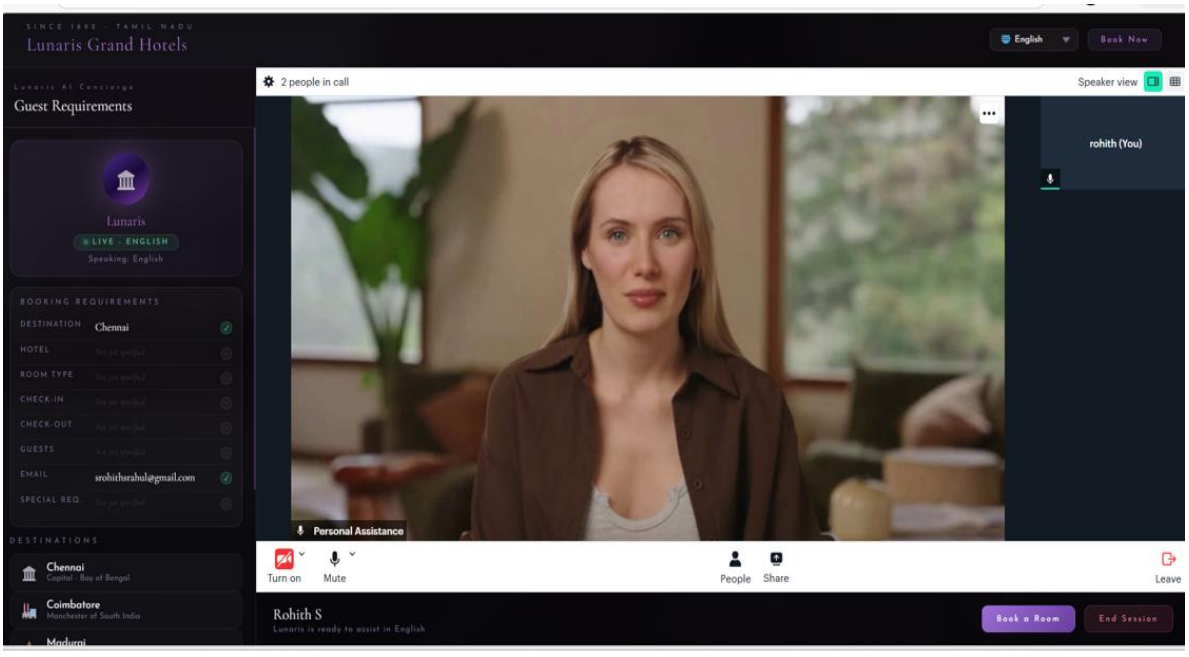

@app.get("/api/health")
async def health():
    return {
        "status": "ok", "version": "3.0.0",
        "service": "Lunaris — Tamil Nadu Grand Hotels AI Concierge",
        "timestamp": datetime.now().isoformat(),
        "destinations": list(HOTELS.keys()),
        "upi_id": UPI_ID,
        "email_configured": bool(SMTP_USER and SMTP_PASS),
```

```
        "services": {
            "livekit": bool(LIVEKIT_URL), "openai": bool(OPENAI_API_KEY),
            "tavus": bool(TAVUS_API_KEY), "n8n_mcp": bool(N8N_MCP_URL)
        }
    }
# ─── Start ──
if __name__ == "__main__":
    logger.info("═" * 62)
    logger.info("  Lunaris — Tamil Nadu Grand Hotels AI Concierge v3.0")
    logger.info("  🌐 Frontend + API: http://localhost:8000")
    logger.info("  📖 API Docs:       http://localhost:8000/docs")
    logger.info("  💳 UPI:            " + UPI_ID)
    logger.info("  📧 Email:          " + ("Configured ✓" if SMTP_USER else "Add SMTP_USER/SMTP_PASS in .env"))
    logger.info("═" * 62)
    uvicorn.run("app:app", host="0.0.0.0", port=8000, reload=True, log_level="info")
```

B. SCREEN SHOTS



Scr.No.1 LOGIN PAGE



Scr.No.2 CUSTOMER INTERACTION PAGE



Scr.No.3 CONFORMATION PAGE

REFERENCES

[1] Shukla R Vernekar, Lava Kumar and Gauri Kalnoor “Hotel Reservation App Integrated with Docker and Jenkins” 2024 International Conference on Emerging Technologies in Computer Science for Interdisciplinary Applications (ICETCS),2024.

[2] Jalel Ktari, Tarek Frikha, Majdi Rekik, Oussema Gassara, Marwen Boujelben and Nader Ben Amor “Decentralized Hotel Reservations: Blockchain Integration for Incentivized Data Sharing and Smart Contracts” 2024 IEEE 4th International Maghreb Meeting of the Conference on Sciences and Techniques of Automatic Control and Computer Engineering (MI-STA),2024.

[3] Josephine D. German, Dave Cullen G. Yap, Guller O. Binoya. “ Design and Development of an Integrated Room Reservation System for Higher Education Institutions”. In 2021 IEEE 8th International Conference on Industrial Engineering and Applications (ICIEA),2021.

[4] Poom Somwong, Siriporn Jaipoonpol, Paskorn Champrasert,Yuthapong Somchit “Smart Room Vacancy Status Checking and Booking System” In 2022 37th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC),2022.

[5] Sinung Suakanto, Riska Yanu Fa'rifah, Widyatasya Agustika Nurtrisha, Tunggul Arief Nugroho “ Development of a Smart Asset Management System for Room Management in Buildings based on QR Code Technology” In 2023 10th International Conference on ICT for Smart Society (ICISS), 2023.

[6] Joseph Wewasse Mahomy, Maria Ioana Oproiu. ” Scalable Webbased Application for Dynamic Multi-Tenant Hotel Reservation Management.” In 2023 15th International Conference on Electronics, Computers and Artificial Intelligence (ECAI),2023.

[7] Wei Wei, Zhengwei Lou. "Design and Implementation of Hotel Room Management System." In 2019 IEEE Symposium Series on Computational Intelligence (SSCI),2019.

[8] David Chunhu Li, Patrick Tran Thi My Ngoc, Tyron Rengulbai, Tamir Zolzaya, Arina Yu-Chen Lu. " MyStay: An Enhanced AI Hotel Reservation System for Seamless Post-Epidemic Hospitality Experience" In 2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS), IEEE, 2021.

[9] M. Murin, F. Jakab and Miroslav Michalko "Advanced reservation system" 2021 19th International Conference on Emerging eLearning Technologies and Applications (ICETA),2021.

[10] Ridho Bramulya Ikhsan, Made Deya Surya Saraswati, Ni Made Pranita , Sulistya Devi, Yudi Fernando, Reagen Yohanes Sayoga and Lianna Wijaya "How To Build Online Hotel Reservation Intention? The Role of Rational Price, Online Review and Trust in Apps" 2022 7th International Conference on Business and Industrial Research (ICBIR),2022.

CONFERENCE

[1] Kiran Kumar .C (611822104020), Kishore.M (611822104021), Rohith.S (611822104043) attended an 3rd International Conference On New Trends In Science. Engineering, Technology And Management at P.S.V.College of Engineering and Technology, Presented a paper entitled "To Implement An Ai Avatar Assistance For Online Hotel Room Booking" on April 2026 (ICNTSETM"26).



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